Traceless-Ricci Descent and Harmonic-Flux Equality

A continuation of the smooth global minimizer problem

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Abstract

For a closed almost-Kähler four-manifold with J-invariant Ricci tensor, we derive an exact first variation of the traceless-Ricci energy from the classical Bach identity of Apostolov–Drăghici. It gives an explicit volume-preserving descending deformation in the nonpositive-scalar range. Combined with the supplied manuscript's harmonic-flux equality theorem, it shows that no non-Bach-flat critical metric can lie in that equality class. This eliminates a candidate construction route for the smooth global minimizer problem; it does not establish smooth attainment.

1 Convention and the classical input

All manifolds below are smooth, closed, connected and four-dimensional. For a Riemannian metric set

$$E = \mathrm{Ric} - \frac{s}{4}g, \quad P(g) = \int_M |E|^2 \, dV_g, \quad S(g) = \int_M s^2 \, dV_g, \quad \mathcal{W}(g) = \int_M |W_{\mathrm{op}}|^2 \, dV_g.$$

The Weyl norm is the curvature-operator norm. Chern–Gauss–Bonnet reads

$$P = 2\mathcal{W} + \frac{S}{12} - 16\pi^2\chi(M). \tag{1}$$

Use the Bach normalization $B = \operatorname{grad}\mathcal{W}$ of [1, §2.3, equation (15)]. This specifies its sign and coefficient; vanishing is independent of that choice. For an almost-Kähler structure (g, J, Ω) with J-invariant Ricci, their Lemma 3 gives

$$B = -\frac{1}{3}(\nabla ds)^{\mathrm{inv}} + \frac{1}{6}(\nabla ds)^{\mathrm{anti}} - \frac{\Delta s}{12}g - \frac{s}{6}E + \mathcal{T}, \tag{2}$$

where $\mathcal{T}$ is symmetric and J-anti-invariant, and $\Delta = d^*d$ is nonnegative. We use this established identity, rather than claim a new formula for the Bach tensor.

The contracted Bianchi identity gives $\operatorname{div} E = ds/4$. Since E is trace-free and J-invariant, integrating (2) against E yields

$$D\mathcal{W}_g[E] = \int_M \langle B, E\rangle \, dV_g = \frac{1}{12}\int_M |ds|^2 \, dV_g - \frac{1}{6}\int_M s|E|^2 \, dV_g. \tag{3}$$

Indeed, the anti-invariant terms are orthogonal to E, and $\int\langle\nabla ds, E\rangle = -\int |ds|^2/4$.

2 A genuine global metric deformation

Theorem 1. *Let (M, g, J, Ω) be closed almost-Kähler, and suppose* $\mathrm{Ric}(JX, JY) = \mathrm{Ric}(X, Y)$. *Then*

$$\boxed{DP_g[E] = \frac{1}{8}\int_M |ds|^2\, dV_g - \frac{1}{2}\int_M s|E|^2\, dV_g.} \tag{4}$$

The smooth global metric path

$$g_t(X, Y) = g(\exp(-tE^\sharp)X, Y) \tag{5}$$

is positive definite and preserves the volume form for every real t. If $s \le 0$, it strictly decreases P for all sufficiently small $t > 0$, unless $ds \equiv 0$ and $sE \equiv 0$.

Proof. For a covariant metric variation h,

$$s'[h] = -\langle \mathrm{Ric}, h\rangle + \operatorname{div}\operatorname{div} h - \Delta_{\mathrm{div}} \operatorname{tr} h, \qquad \Delta_{\mathrm{div}} = \operatorname{div}\nabla = -\Delta.$$

For $h = E$, this becomes $s'[E] = -|E|^2 + \Delta_{\mathrm{div}} s/4$, while $dV'[E] = 0$. Thus

$$DS_g[E] = -2\int_M s|E|^2\, dV_g - \frac{1}{2}\int_M |ds|^2\, dV_g. \tag{6}$$

Combine (1), (3) and (6) to obtain (4). The endomorphism $E^\sharp$ is self-adjoint and trace-free. Consequently $\exp(-tE^\sharp)$ is positive definite with determinant one. Hence (5) is an actual metric path, not a formal curvature variation, and has fixed volume form. Its derivative at zero is $-E$, giving

$$\left.\frac{d}{dt}P(g_t)\right|_{t=0} = -\frac{1}{8}\int_M |ds|^2\, dV_g + \frac{1}{2}\int_M s|E|^2\, dV_g. \tag{7}$$

Both terms are nonpositive when $s \le 0$. Equality requires both $ds = 0$ and $s|E|^2 = 0$; connectedness gives the stated exceptions. □

The varied metrics need not remain almost-Kähler. This is appropriate for an infimum over *all* smooth metrics, including a unit-volume normalization. In particular the deformation is not a claim about a minimum restricted to a fixed symplectic form.

Corollary 2. *For a constant-scalar metric in Theorem 1,*

$$DP_g[E] = -\frac{s}{2}P(g).$$

If $s \neq 0$ and $P(g) > 0$, the path $g(\exp(t\operatorname{sgn}(s)E^\sharp)\cdot, \cdot)$ has derivative $-|s|P(g)/2 < 0$. Thus a constant-nonzero-scalar P-critical metric in this entire almost-Kähler, J-invariant-Ricci class is Einstein.

If $s < 0$, one may also conformally correct this path to retain the initial constant scalar curvature. The linearization of the equation $6\Delta_g u + s_g u = s_0 u^3$ at $u = 1$, $s_g = s_0 < 0$, is $6\Delta_g - 2s_0$, which is invertible. The implicit function theorem gives the correction. Since the conformal first variation of P vanishes at a constant-scalar metric, the strict derivative is unchanged.

Corollary 3. *If $s \le -\sigma < 0$ and $P > 0$ in Theorem 1, then*

$$\|\operatorname{grad} P\|_{L^2} \ge \frac{\|ds\|_{L^2}^2}{8\sqrt{P}} + \frac{\sigma}{2}\sqrt{P}.$$

Proof. Apply Cauchy–Schwarz to $DP[E] = \int\langle \operatorname{grad} P, E\rangle$ and use (4) and $\|E\|_2 = \sqrt{P}$. □

This estimate concerns the exact geometric class in the theorem. It does not supply the derivative control needed to extend the identity to arbitrary nearly aligned, degenerating metrics.

3 Consequence for the supplied harmonic-flux decomposition

Write $\mathcal{B} : \Lambda^+ \to \Lambda^-$ for the off-diagonal curvature block, so $P = 4\|\mathcal{B}\|_2^2$. For a nonzero harmonic anti-self-dual two-form h, put

$$H = \|h\|_2^2, \qquad F_h = \sqrt{2}\int_M |\mathcal{B}^* h|\, dV_g, \qquad L_h = \frac{P/2 - F_h^2/H}{4\pi^2} \geq 0.$$

The supplied continuation [2, Sections 4–5] proves the following equality statement, including removal across the harmonic zero set:

> At constant scalar curvature, $L_h = 0$ implies that g is Einstein or admits a smooth global almost-Kähler structure with J-invariant Ricci.

This is the precise additional input used in the next proposition.

Proposition 4. *Assuming the equality statement above, a smooth closed P-critical metric with $L_h = 0$ for some nonzero harmonic anti-self-dual h is Bach-flat. If its scalar curvature is nonzero, it is Einstein.*

Proof. For $g_t = e^{2tf} g$, conformal invariance of $\mathcal{W}$ and (1) give

$$DP_g[2fg] = \int_M f \Delta s\, dV_g.$$

Criticality implies that s is constant. If $s \neq 0$, the equality statement and Corollary 2 force $E = 0$. If $s = 0$, the first variation of S vanishes, and (1) gives $\operatorname{grad} P = 2B$. Criticality then implies $B = 0$. □

Therefore, on the negative-Yamabe candidates with $P_* > 0$ already identified in the project, a hypothetical smooth global minimizer must have

$$L_h(g) > 0 \qquad \text{for every nonzero } h \in \mathcal{H}_g^-. \tag{8}$$

At a fixed such metric with $b_- > 0$, compactness of the harmonic unit sphere and continuity give $\min_{\|h\|_2=1} L_h(g) > 0$. This last positive number depends on the metric. It is not a uniform bound along a minimizing sequence.

The direction of this consequence matters: forcing $L_h = 0$ would *exclude* the desired non-Bach-flat minimizer. It would not establish its existence. A sharp inequality attained by such a minimizer must accommodate a nonzero harmonic-alignment remainder or a different geometric structure.

4 Two normalization checks and the remaining problem

For a product of constant-curvature surfaces with Gaussian curvatures a, b and product volume V, one has $s = 2(a+b)$ and $P = (a-b)^2 V$. The eigenvalues of $E^\sharp$ on the two factors are $(a-b)/2$ and $-(a-b)/2$. Under $g_t = g(\exp(tE^\sharp)\cdot, \cdot)$,

$$P(g_t) = V\left(ae^{-t(a-b)/2} - be^{t(a-b)/2}\right)^2.$$

Its derivative is $-(a+b)(a-b)^2 V = -sP/2$, independently checking the coefficient and sign of Corollary 2.

There is also a direct distinction between the two global variational problems in the supplied manuscript. Let

$$R = \int |\operatorname{Ric}|^2 = P + S/4, \quad T_s = \int s, \quad V = \operatorname{Vol}(M), \quad \mathcal{E} = 4 - \frac{T_s^2}{VR}.$$

At a smooth P-critical metric with nonzero constant scalar curvature, the first variation of $S - T_s^2/V$ vanishes. Thus

$$\boxed{D\mathcal{E}_g[E] = \frac{2sP^2}{R^2}.} \tag{9}$$

Indeed, $\mathcal{E} = 4P/R$ to first order there, $DP = 0$, and $DS[E] = -2sP$. If $P > 0$, (9) is nonzero. Hence smooth attainment of the normalized defect does not produce the sought non-Bach-flat P-minimizer.

What remains. The target is still a smooth global minimizer of P on a closed manifold which is not Bach-flat. No smooth-attainment theorem or such metric is constructed here. The result proved here is the explicit descending deformation and the consequent exclusion of the supplied flux-equality class. The Bach identity is classical; the connection to the project's flux equality is the continuation recorded in this note.

References

[1] V. Apostolov and T. Drăghici, *Almost Kähler 4-manifolds with J-invariant Ricci tensor and special Weyl tensor*, Quart. J. Math. 51 (2000), 275–294. arXiv:math/9911196. The input is §2.3, equation (15), Lemma 3.

[2] Supplied project manuscript, *Einstein Gaps on Four-Manifolds: Subsequent Mathematical Developments*, 8 October 2026, `03_Subsequent_Mathematical_Developments.tex`, Sections 4–5 (harmonic flux, equality and removability).